# ShieldMate — Testing Plan

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Every increment in the Development Plan ends at a test gate with the same number: Increment 1 is done when Gate T1 passes. Automated tests from earlier gates keep running, so nothing that worked stops working.

## Approach

Six levels of testing, from fast automated checks to the timed rehearsal. Automate what you can, because there is no time to repeat manual tests at every gate.

| Level | Tool | Checks | Written by | Run |
|---|---|---|---|---|
| Unit | pytest | One module's logic: filtering, state machine, scoring, hashing, guards | Module owner, with the code | Before every merge |
| API | pytest + FastAPI TestClient | Routes against the contract: status codes, response shapes, error cases | Route owner | Before every merge |
| Contract | pytest, written in Increment 0 | Every route's response matches the frozen schema | Kishaann | Before every merge |
| UI flow | Playwright, several browser contexts as different people | Multi-device flows: alert to phone, drill sync | Abhiram | At every gate |
| Device | Two real phones + laptop on the hotspot | What automation can't show: real phones, Wi-Fi, fonts, touch | Whole team | At every gate |
| Lab | Wazuh lab + attack script | Live detection and blocking | Ishaan | Gate T5 only |

How a gate runs:

1. Everyone merges to main.
2. `make test` runs every automated test from **all** gates so far. All must pass.
3. The gate's manual tests run on the demo laptop and two phones.

4. Tick the gate's sign-off box below. If any test fails, the increment isn't done; the cut rules in the Development Plan apply.

Test IDs read as gate and number: **T2.05** is the fifth test of Gate T2, which belongs to Increment 2.

## Environments and test data

| Environment | What it is | Used for |
|---|---|---|
| Dev laptops | Each person's laptop, stubs or their own branch | Unit, API and contract tests |
| Demo rig | The presenter's laptop as server and hotspot, one Android phone (Chrome), one iPhone (Safari) | UI flow and device tests at every gate; rehearsals |
| Lab | Ishaan's laptop: Wazuh manager, Windows VM with agent, attack script on a second machine | Gate T5 |

All tests use the same fixed data, so a result means the same thing on every machine:

- **Clinic seed** ( `seed/clinic.json` ): Sunrise Family Clinic, 3 staff, 3 devices, backup limit 24 h, never-block list.
- **Replay file** (Design doc, Appendix B): 5 records, including the poisoned username and two filtered events.
- **Scenario file** (Design doc, Appendix A): 4 steps with role-split options.
- **Synthetic backup**: regenerated by Reset; 50 files, 2 altered, 9 days old, always connected.
- **Good backup fixture** (tests only): same generator with no faults, to prove a pass is possible.
- **Fake Wazuh client**: records every call instead of sending it. Used by every test except Gate T5, so tests can prove nothing was sent.
- **Fake clock**: lets the 5-minute timeout and 1-hour expiry run in milliseconds.

# Gate T0: Foundations (H2)

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T0.01 | Schema on an empty database | Unit | All tables exist, including live-mode ones | Kishaann |
| T0.02 | Every contract route | Contract | Each returns 200 and fixture data matching its schema | Kishaann |
| T0.03 | Error shape | API | An invalid request returns 4xx with `{error, message}` | Kishaann |
| T0.04 | Scenario file | Unit | Loads; 4 steps; every role with options has exactly one correct option per step | Cetas |
| T0.05 | Replay file | Unit | Loads; 5 records; click numbers as designed | Ishaan |
| T0.06 | Backup generator | Unit | 50 originals; manifest hashes match them; exactly 2 backup files differ; age 9 days | Cetas |
| T0.07 | Reset | API | Empty database becomes the seeded clinic; running it twice gives the same state | Kishaann |
| T0.08 | Start on the demo rig | Device | `./start.sh` runs; both phones on the hotspot open the start page | Abhiram |

- [ ] Gate T0 passed

# Gate T1: Basic working product (H8)

The most important gate: if T1 passes, you have a demo.

## Alerts

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T1.01 | Replay the password attack | API | Front desk gets 1 card; its text names RECEPTION-PC and "8+" | Kishaann |
| T1.02 | Tap Done | API | Alert is done; the owner gets a notification | Kishaann |

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T1.03 | Replay the ransomware attack | API | Owner gets the ransomware card | Kishaann |
| T1.04 | Role wording | API | Front desk and owner get different steps for the same alert | Kishaann |

## Drill

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T1.05 | Create and join | API | Code is 4 digits and unique among open drills; join returns a player ID; a wrong code is rejected | Cetas |
| T1.06 | State machine | Unit | Lobby, open, review, open … finished, in that order only; only the owner can advance | Cetas |
| T1.07 | Answer once | API | A second answer to the same step is rejected; the score doesn't change | Cetas |
| T1.08 | Late answer | API | An answer after the step closes gets `step_closed` | Cetas |
| T1.09 | Role options | API | Staff see staff options only; the owner sees owner options; a role with none sees a waiting message | Cetas |
| T1.10 | Step closes | API | When all players have answered, the step moves to review | Cetas |
| T1.11 | Scoring | Unit | Scripted run (Meena all right, Ravi wrong at steps 1 and 2) gives the expected score per stage | Cetas |

## Restore check

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T1.12 | Demo backup | Unit | FAIL with all three reasons: 9 days old, 48 of 50 match, always connected | Cetas |
| T1.13 | Good backup | Unit | PASS | Cetas |
| T1.14 | Writes only to the sandbox | Unit | Files outside `restore_sandbox/` are unchanged after a run | Cetas |

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T1.15 | Before the first run | API | After Reset, the latest result is `NOT_TESTED` | Cetas |
| T1.16 | Drill trigger | API | At step 3, the owner's right answer shows Run restore check | Cetas |

### End to end

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T1.17 | Alert reaches the phone | UI flow | Card appears on Meena's screen within 3 s of the replay; Done reaches the owner within 3 s | Abhiram |
| T1.18 | Drill stays in sync | UI flow | Three players finish the drill; all show the same step within 3 s of the owner advancing | Abhiram |
| T1.19 | **Full demo story** | Device | Run on 2 real phones and the laptop, twice, with Reset in between; no restart needed | All |

- [ ] Gate T1 passed: basic working product

## Gate T2: Demo-ready core (H12)

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T2.01 | Noise filter | Unit | Level below 7, or no lesson, is filtered; the hidden count is right | Kishaann |
| T2.02 | Grouping | Unit | Same rule and PC within 10 min bumps the count (8+ to 12+); after 10 min a new card appears | Kishaann |
| T2.03 | Poisoned username shown exactly | UI flow | The From-the-log box shows the username character for character | Abhiram |
| T2.04 | Log text can't run as code | UI flow | A username containing `<script>` or HTML shows as plain text and runs nothing | Abhiram |

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T2.05 | Languages | Device | Toggle switches the alert card on both phones; Kannada and Hindi letters render; a fluent speaker signs off the wording | Abhiram |
| T2.06 | Impact meter | Unit | Each step's device states give the services in the design doc table | Cetas |
| T2.07 | End-of-drill results | API | Missed action IDs match the scripted run | Cetas |
| T2.08 | Single point of failure | API | The owner screen data flags that only Dr. Rao can restore | Cetas |
| T2.09 | Phone reloads mid-drill | Device | Other players unaffected; the reloaded phone rejoins | All |
| T2.10 | Phone briefly offline | Device | 10 s with Wi-Fi off: screen keeps the last data, then catches up | All |
| T2.11 | Matches the wireframes | Device | Each screen on Android and iPhone: layout as designed, buttons at least 44 px, nothing cut off | Abhiram |
| T2.12 | Honesty badges | UI flow | Replay mode and Simulated backup badges show wherever they apply | Abhiram |
| T2.13 | Earlier gates | All | T0 and T1 automated tests pass; T1.19 run passes again | All |

- [ ] Gate T2 passed

## Gate T3: Guarded AI rewording (H14)

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T3.01 | Log text never reaches the AI | Unit | With the poisoned username in the alert, a spy on the provider sees only the headline and why-text | Kishaann |
| T3.02 | Guard rejects bad output | Unit | Output that changes a number, adds digits or a link, or runs over 200 characters falls back | Kishaann |

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T3.03 | No provider | Unit | Returns the fallback wording with `source: "fallback"` | Kishaann |
| T3.04 | Steps never change | UI flow | Steps and the log box are identical with the toggle on and off | Abhiram |
| T3.05 | Flag off | API | With `AI_REWORD` off, the toggle is hidden and the route returns 404 | Kishaann |

- [ ] Gate T3 passed, or Increment 3 cut and its flag left off

## Gate T4: Approval flow, simulated (H16)

Every T4 test runs with the fake Wazuh client and checks it received **no** calls.

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T4.01 | Proposal | Unit | The password alert proposes `BLOCK_IP` for its source IP; the ransomware alert proposes nothing | Kishaann |
| T4.02 | Target checks | Unit | Refused: never-block list, the laptop, the Wazuh server, invalid IPv4, IPv6, empty or missing IP. Each falls back to advice only | Kishaann |
| T4.03 | Approve | API | Status `executed` with "Simulated"; decided by the owner via tap; logged | Kishaann |
| T4.04 | Decline | API | Status `declined`; logged; nothing executed | Kishaann |
| T4.05 | Approve twice, or after decline | API | Second request rejected; state unchanged | Kishaann |
| T4.06 | Timeout, standing rule on | Unit, fake clock | After 5 min: `executed` via `standing_rule`; owner notified | Kishaann |
| T4.07 | Timeout, standing rule off | Unit, fake clock | After 5 min: not executed; backup approver asked | Kishaann |
| T4.08 | Expiry | Unit, fake clock | After 1 hour the block is `expired` and the device shows OK | Kishaann |

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T4.09 | Standing-rule setting | API | Turning it on or off records who and when | Kishaann |
| T4.10 | Device strip | API | RECEPTION-PC goes OK, Under attack, Protected as the flow runs | Cetas |
| T4.11 | Drill can't act | API | Choosing a `BLOCK_IP` drill option creates no action | Cetas |
| T4.12 | Owner screen | UI flow | Approval card shows pending, approved and declined states; countdown; log entries in order | Abhiram |
| T4.13 | Flag off | API | With `ACTIONS` off, no proposals are made and no approval cards appear | Kishaann |
| T4.14 | Earlier gates | All | All earlier automated tests pass; T1.19 run passes with the approval step added | All |

- [ ] Gate T4 passed

# Gate T5: Live Wazuh (H19)

## Lab gate (H14)

These two decide whether Increment 5 goes ahead at all.

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T5.01 | Detection | Lab | The attack script triggers rule 60204, and the alert in the Wazuh dashboard carries the attacker's source IP | Ishaan |
| T5.02 | Manual block | Lab | A block sent by hand through the Wazuh API stops the script's attempts | Ishaan |

## Gate T5

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T5.03 | Same shape as replay | Integration | A live alert, normalised, matches the replay record field for field (except time and counts) | Ishaan |
| T5.04 | Attack to card | Lab | 5 runs: card on the phone within 30 s every time; record each time | Cetas |
| T5.05 | Approve to block | Lab | 5 runs: attack stops within 30 s of Approve; firewall rule visible on the VM | Cetas |
| T5.06 | Expiry | Lab | With a 2-minute test expiry, the block lifts by itself and the attack gets through again | Ishaan |
| T5.07 | Undo | Lab | Undo removes the block. **If it fails, the Undo button is replaced by "Lifts at …"** | Ishaan |
| T5.08 | Never-block in live | Lab | An attack from a never-block address produces advice only; the Wazuh API receives no call | Ishaan |
| T5.09 | Fallback while live | Device | With the lab switched off, Backstage replay still produces cards; no restart needed | Ishaan |
| T5.10 | Credentials stay on the server | API | No Wazuh password or token appears in any API response or in the built frontend files | Kishaann |
| T5.11 | Earlier gates | All | All earlier automated tests pass, still using the fake client | All |

- [ ] Lab gate passed at H14, or Increment 5 skipped
- [ ] Gate T5 passed

# Gate T6: Freeze and rehearse (H19–H24)

| ID | Test | Level | Passes when | Owner |
|---|---|---|---|---|
| T6.01 | Full suite on the demo laptop | All | Every automated test passes on the machine you will present from | Kishaann |
| T6.02 | Timed run, normal | Rehearsal | Full story in 5:00 or less | All |
| T6.03 | Timed run, internet off | Rehearsal | Same story with the laptop's internet disconnected; nothing breaks | All |
| T6.04 | Timed run, phone fails | Rehearsal | One phone dies mid-drill; Backstage opens that player on a laptop and the drill finishes | All |
| T6.05 | Lab off (if live mode is in) | Rehearsal | The live scene switches to replay with "Simulated action" and the story still lands | Ishaan |
| T6.06 | Backup video | Device | The recorded run plays offline from the demo laptop | Abhiram |

- [ ] Gate T6 passed

## Pre-flight checklist (just before going on stage)

- [ ] Demo laptop and both phones charged, or on power
- [ ] Laptop hotspot on; both phones joined; ShieldMate open on each, with the right person picked
- [ ] Do not disturb on, on all three devices; screen lock off on the phones
- [ ] Backstage: Reset demo pressed; replay buttons show click 1
- [ ] Projector mirrors the laptop; browser zoom checked from the back of the room
- [ ] Live mode only: lab VM running, agent Active, attack script ready, credentials loaded
- [ ] Backup video open in a player, paused at the start

# Bugs

Log every bug as a GitHub issue with its test ID, its severity and who found it.

| Severity | Meaning | Example | Fix when |
|---|---|---|---|
| S1 | Breaks the demo story | Phones stop syncing; restore check crashes | Immediately, by whoever owns the module; others help |
| S2 | Visible on stage, story still works | A label cut off on iPhone; wrong count on a card | Before the next gate; after H19 only if the fix is small and tested |
| S3 | Not visible in the demo | Edge case in an error message | After the demo |

After the H19 freeze, only S1 and small S2 fixes merge, and each one re-runs `make test` and the full demo story before merging.